# COHERENCE, CORRESPONDENCE, CALIBRATION: A META-EPISTEMOLOGY APPLIED TO AUSTRIAN ECONOMICS

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# Introduction

- Markets generate abundant signals — yet economic frameworks struggle to adapt under persistent anomaly.
- Why do such information-rich systems still experience instability?
- Do systems fail because they coordinate signals but lack an explicit calibration axis to secure metric integrity over time?
- Many intellectual frameworks privilege one reasoning mode while leaving others implicit.
  - Deductive systems risk circularity
  - Empiricism risks noise
  - Adaptive heuristics risk drift
- This work argues: complex adaptive systems require coherence, correspondence, and an explicit calibration axis.

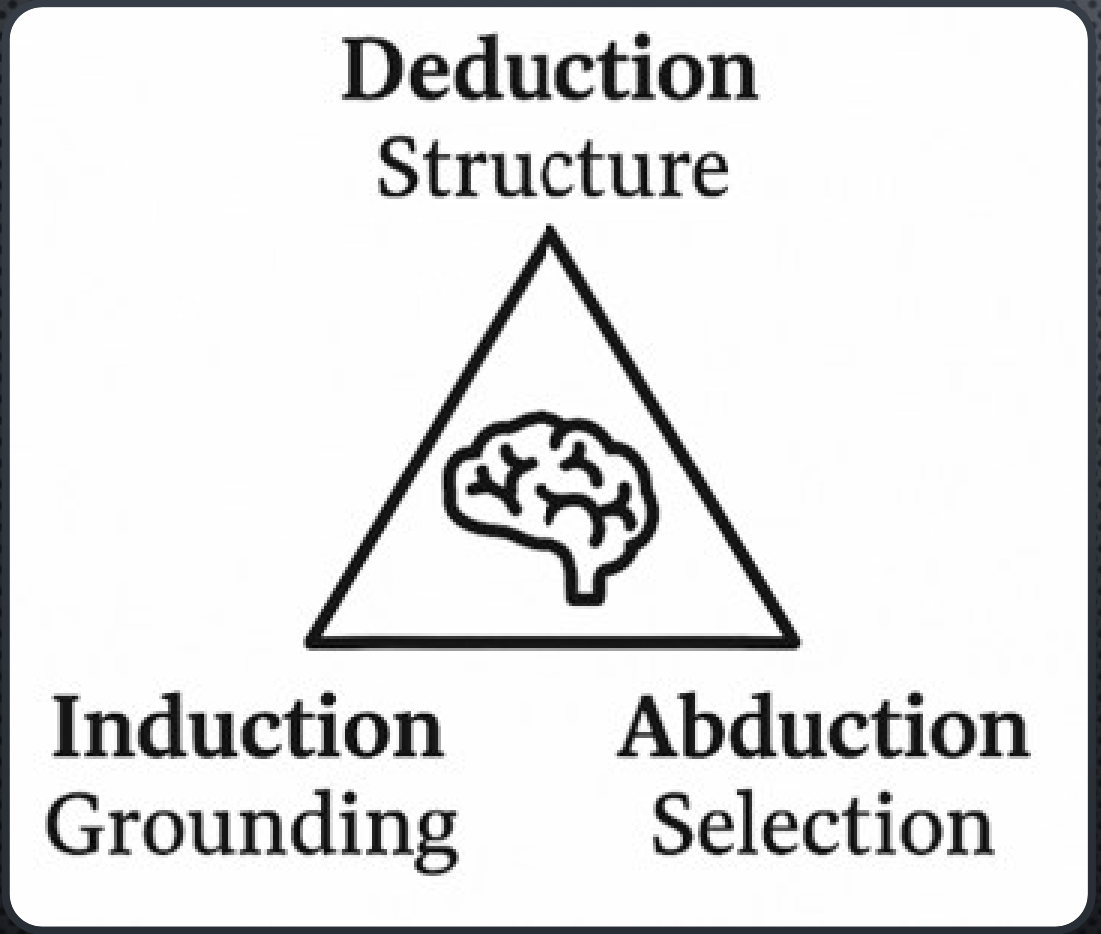
# This presentation:

- introduces the **Coherence–Correspondence–Calibration (CCC)** Framework as a structural architecture for evaluating adaptive systems
- applies it to Austrian Economics as a structural case study.

CCC Framework → Analysis → Implication → Working Example

Note: The critique is grounded in system architecture and control-theoretic design principles, not doctrinal disputes.

# THREE MODES OF REASONING




**KEY INSIGHT:** THESE THREE MODES JOINTLY FORM THE MINIMAL ARCHITECTURE OF ADAPTIVE INTELLIGENCE AND KNOWLEDGE SYSTEMS.

# COHERENCE–CORRESPONDENCE–CALIBRATION FRAMEWORK

**CCC FRAMEWORK** ASSESSES THE WORLDVIEW ON THE FOLLOWING THREE CRITERIA:

1. ARE ITS RULES AND INTERNAL LOGIC STRUCTURALLY CONSISTENT? → **COHERENCE (DEDUCTION)**
2. DO ITS SIGNALS RELIABLY TRACK REAL-WORLD BEHAVIOUR? → **CORRESPONDENCE (INDUCTION)**
3. DOES IT CONTAIN AN EXPLICIT MECHANISM TO SECURE AND RECALIBRATE ITS EVALUATIVE STANDARD OVER TIME? → **CALIBRATION (ABDUCTION)**

THE CCC TRIAD IS DIRECTLY MERGED WITH THE THREE CLASSICAL MODES OF REASONING.

# AVIATION ANALOGY

Coherence → engineering design rules

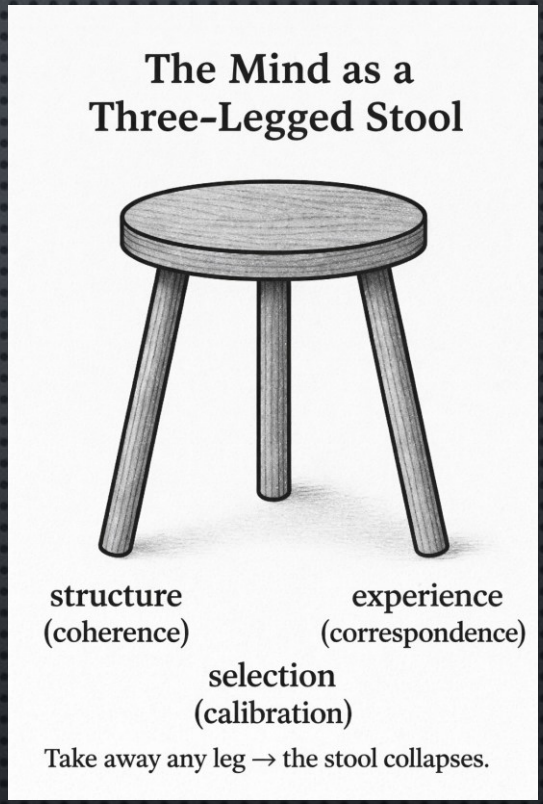
Correspondence → sensor feedback

Calibration → periodic instrument calibration

# IRREDUCIBILITY OF THE THREE MODES

- Deduction requires premises → premises require inductive grounding.
- Induction requires categories → but categories themselves require logical structure.
- Both require hypothesis selection under uncertainty → abduction.

→ No mode can justify its own starting point.
→ isolation collapses function.



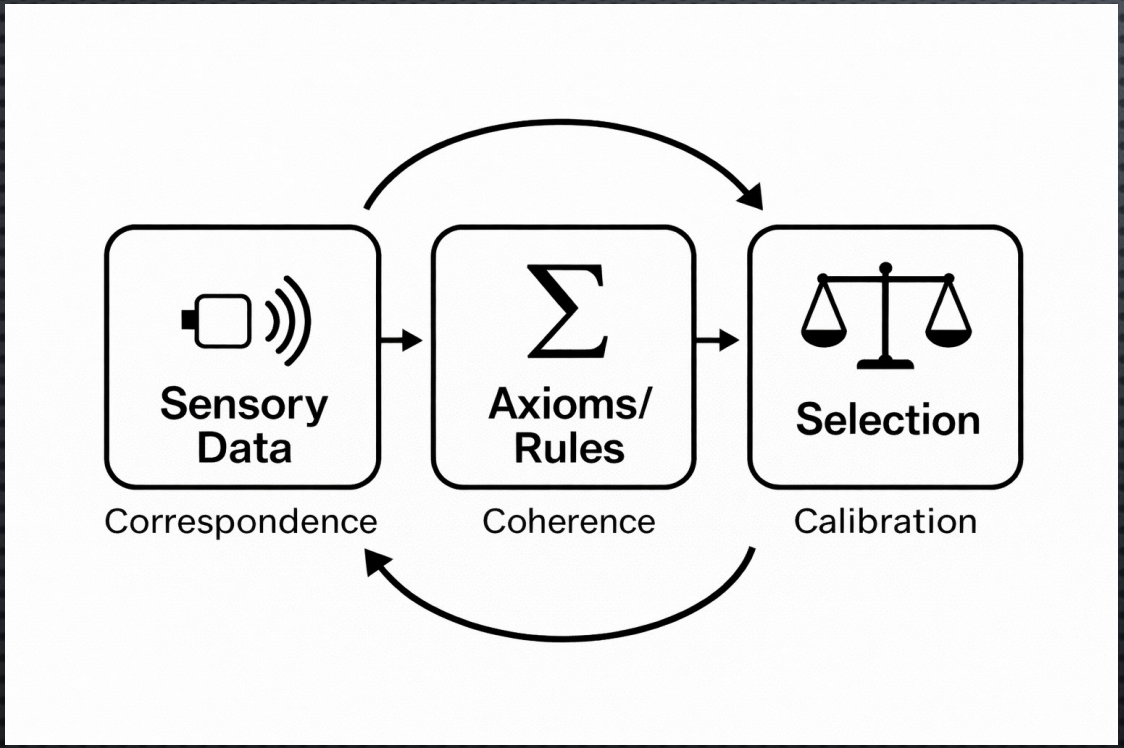

**Key Insight:** Intelligibility requires the triadic reasoning architecture.[1]

- deduction ↔ induction ↔ abduction
- Intelligence = closed reasoning loop

[1] Charles Sanders Peirce, *Collected Papers*, 5.171 (c.1903)

# AI SYSTEMS AS AN ANALOGY

LLMs make visible, at scale, the interaction of all three reasoning modes. Their failure modes expose the necessity of each axis.




- Without induction (correspondence), the system cannot generalize — it collapses into empty formalism;
- Without deduction (coherence), it produces noise;
- Without abduction (calibration), — it loses error-correction and produces confident but ungrounded output.

**Key Insight:** A single mode alone is non-generative — the system cannot produce intelligence. Any economic system that claims to be adaptive must instantiate all three structurally.

# CASE STUDY: AUSTRIAN ECONOMICS — EPISTEMIC STRUCTURE

- **Foundational Theory Layer**
  - Mises — Purposeful human action as foundational axiom (Praxeology)
  - A priori deductive system (informed by Neo-Kantian tradition); Explains intelligibility, not prediction
  - Value-neutral in scope
- **Market Process Layer**
  - Hayek — dispersed knowledge, learning, adaptation (Induction)
  - Kirzner — Profit/Loss as error, discovery, correction (Abduction)

**Key Insight:**

- AE distributes learning across market actors, but insulates the theory describing that learning from the same revision process.
- The axiomatic base remains methodologically insulated from empirical revision.

# CCC FRAMEWORK ANALYSIS —ARCHITECTURAL ASYMMETRY IN AE

| CCC Axis | Level | AE |
|---|---|---|
| Coherence | Theory | **Very high**—axiomatic derivation from action |
| Correspondence | Market Process | **Partial / operational** — prices and entrepreneurial error-correction exist but do not function as foundational constraints on theory. |
| Calibration | System governance | **Underspecified** — Foundational axioms treated as fixed; no explicit mechanism for revising system-level outcomes under persistent theory–reality divergence. |

**Key Insights:**

- Intelligibility at the theoretical level cannot arise from deduction alone if adaptive learning occurs at the system level.
- A system describing adaptive intelligence must itself be calibratable. Otherwise, the model can remain internally coherent while reality diverges indefinitely.

# Methodological choice and its consequence

- Treating economics as a purely a priori science is a legitimate methodological choice. That choice prioritizes logical necessity.
- Deduction organizes reasoning, but induction and abduction are required to ground and revise it.

The structural consequence is this:

- No governed mechanism for premise-level revision under persistent anomaly.
- No formal procedure for distinguishing epistemic limitation from real-world distortion.
- No layer asks whether the evaluative criteria themselves remain stable under compounding dynamics?
- A system without formal meta-calibration can remain internally coherent while accumulating structural drift under its own success conditions.

# HUMEAN IS-OUGHT NORMATIVE BRIDGE

- Descriptive statements about the world (what **is**) do not logically entail normative conclusions (what **ought** to be).
- IS claim about how markets behave → OUGHT claim about what policy should do
- OUGHT requires a normative bridge; Therefore: every policy claim embeds a VALUE premise.

Implicit

Both economic traditions show symmetry in the **IS-VALUE-OUGHT** architecture

**Austrian Economics**

- Markets coordinate dispersed knowledge (IS)
- Efficiency / voluntary exchange is preferable (evaluative norm)
- Therefore: non-intervention (OUGHT)

**Keynes/MMT (As a Reference)**

- Markets produce instability / underemployment (IS)
- Stability / full employment is preferable (evaluative norm)
- Therefore: intervention (OUGHT)

**Key Takeaway**: Normativity is unavoidable. The question is whether the evaluative premise is explicit and structurally constrained under feedback.

# METRIC TARGETING → DRIFT

**GOODHART'S LAW:** WHEN A MEASURE BECOMES A TARGET, IT CEASES TO BE A GOOD MEASURE

**ELEVATE PROFIT/OUTPUT INTO A DE-FACTO EPISTEMIC AUTHORITY.**

- WHEN METRIC BECOMES THE GOAL → BEHAVIOR SHIFTS TO GAME THE METRIC
- AUSTRIANS: METRIC DOMINANCE THROUGH MARKET PROFIT SIGNAL → VULNERABLE TO DRIFT
- KEYNES: METRIC DOMINANCE THROUGH POLICY OPTIMIZATION → VULNERABLE TO DRIFT
- EVALUATIVE PREMISES FUNCTION AS FIRST-ORDER SELECTION METRICS IN RECURSIVE SYSTEMS.

- HUME (NORMATIVITY IS UNAVOIDABLE)
- GOODHART (METRICS DRIFT UNDER OPTIMIZATION)
- RECURSIVE SYSTEMS THEORY(FEEDBACK AMPLIFIES DRIFT)

} SYSTEMS-LEVEL SYNTHESIS

**KEY INSIGHT:** RECURSIVE EVALUATIVE SYSTEMS REQUIRE SECOND-ORDER GOVERNANCE.

# Cattalaxy and the Calibration Question

- **Catallaxy** — solves the knowledge aggregation problem.
- Relies on decentralized price feedback to coordinate action.
- price/profit → signal. But what exactly does the signal measure?
- Profit (proxy) measures success within current exchange conditions at current prices, not system coordination of capital structures across time
- Therefore, max(profit as proxy) ≠ max(coordination)
- Optimization pressure shifts behavior toward maximizing the proxy
- Behavior shifts from "produce coordination → profit follows" to "maximize profit → regardless of mechanism."

**If calibration is necessary, where could it exist?**

# Digital Catallaxy in Practice — CCC in Decentralized AI Markets

- **Bittensor** — an intelligence economy structured around three actor classes:
  - **Miners** — Producers of Intelligence
  - **Validators** — Empirical Evaluators
  - **Subnet Owners** — Incentive Architects
- The **Protocol Layer** has all the rules, constraints, consensus logic (OUGHT)
- token emission venture funding for decentralized experimentation

**CCC Axes:**

- **Coherence**: protocol rules and consensus logic
- **Correspondence**: empirical scoring of outputs
- **Calibration**: rubric evolution, subnet competition, incentive realignment via Yuma consensus

**Key Insight**: This illustrates how decentralized systems can embed metric governance and adaptive calibration into protocol logic.

# DIGITAL CATALLAXY: WITH AND WITHOUT META-CALIBRATION

| Dimension | Spontaneous Order only | Spontaneous Order + Calibration |
|---|---|---|
| Value Definition | Value = whatever the incentive function currently rewards | Value is periodically reassessed through rubric evolution and rule revision |
| Metric Dynamics | Optimization pressure leads to metric gaming (Goodhart drift) | Metric governance adjusts scoring rules to preserve signal integrity |
| Information Asymmetry | Technical advantage compounds; influence concentrates | Validator influence and incentives recalibrated to mitigate concentration |
| Equilibrium Outcome | Predatory or low-quality equilibria can stabilize endogenously | Extractive strategies detected and structurally disincentivized |
| Normative Authority | No explicit evaluative standard; normativity emerges implicitly through outcomes | Evaluative criteria (OUGHT) encoded explicitly in protocol logic (Yuma) and governance |

**Key Insight:** Coordination alone optimizes within rules; calibration governs the evolution of the rules.

# Conclusion: What CCC Reveals

- All adaptive intelligence systems require CCC triad and reasoning modes
- Austrian economics achieves high coherence; It operationalizes correspondence through price signals; Calibration operates locally (agent-level).
- No formal mechanism governs premise-level revision.
- Over long horizons, systems that lack governed recalibration risk preserving internal consistency while accumulating external distortion.
- Hume + Goodhart + Recursive systems logic implies necessity of second-order governance
- CCC formalizes spontaneous order into an explicitly adaptive market architecture

**Key takeaways:**

- Digital systems now demonstrate that CCC-integrated markets are structurally possible.
- Coordination without calibration drifts; calibration without coordination stagnates. Intelligent architecture requires both.

# Q&A

cccframework.ca

# APPENDIX: ECONOMIC CONTROL SYSTEM ARCHITECTURE

| System | Calibration Type | Control structure |
|---|---|---|
| Austrian | Market-endogenous error correction | decentralized feedback |
| Keynesian | Institutional calibration | centralized controller |
| Bittensor | Multi-layer endogenous + protocol governance | layered decentralized controllers |

**Key insight:** complex recursive systems require both decentralized knowledge and explicit metric governance.

# APPENDIX: MULTI-LAYER CALIBRATION ARCHITECTURE IN BITTENSOR

| Layer | Actors | What Competes | Selection Signal | Calibration Mechanism |
|---|---|---|---|---|
| **1. Compute / Model Layer** | Miners | AI models & inference strategies | Validator scores | Reward allocation (weights from Yuma) |
| **2. Evaluation Layer** | Validators | Scoring accuracy & evaluation methods | Weight consensus / validator alignment | Validator reward / influence adjustment |
| **3. Capital Allocation Layer** | Delegators / TAO holders | Validators (who evaluates best) | Staking flows | Delegation shifts based on performance |
| **4. Intelligence Market Layer** | Subnets | Task domains & evaluation systems | Subnet token value / emissions | Stake flows and subnet competition |
| **5. Constitutional Layer (protocol governance)** | Protocol governance / TAO governance | Protocol rules & system parameters | Governance voting | Runtime upgrades / parameter changes |